In Closed Loop ABDA,

$$I_1 + 5I_3 - 4I_2 = 0$$
$$I_1 - 4I_2 + 5I_3 = 0 \qquad (1)$$

Closed Loop BCDB

$$2(I_1 - I_3) - 3(I_2 + I_3) - 5I_3 = 0$$
$$2I_1 - 2I_3 - 3I_2 - 3I_3 - 5I_3 = 0$$
$$2I_1 - 3I_2 - 10I_3 = 0 \qquad (2)$$

Closed Loop ABCA

$$I_1 + 2(I_1 - I_3) + (I_1 + I_2) - 2=0$$
$$I_1 + 2I_1 - 2I_3 + I_1 + I_2 = 2$$
$$4I_1 + I_2 - 2I_3 = 2$$

$$\begin{bmatrix} 1 & -4 & 5 \\ 2 & -3 & -10 \\ 4 & 1 & -2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 1 & -4 & 5 \\ 2 & -3 & -10 \\ 4 & 1 & -2 \end{bmatrix}$$
$$= 1(6+10) + 4(-4+40) + 5(2+12)$$
$$= 16 + 144 + 70 = 230.$$

$$\Delta_1 = \begin{bmatrix} 0 & -4 & 5 \\ 0 & -3 & -10 \\ 2 & 1 & -2 \end{bmatrix}$$
$$= 0(6+10) + 4(0+20) + 5(0+6)$$
$$= 80 + 30 = 110$$

$$\Delta_2 = \begin{bmatrix} 1 & 0 & 5 \\ 2 & 0 & -10 \\ 4 & 2 & -2 \end{bmatrix}$$
$$= 1(0+20) + 0(-4+40) + 5(4+0)$$
$$= 20 + 20 = 40$$

$$\Delta_3 = \begin{bmatrix} 1 & -4 & 0 \\ 2 & -3 & 0 \\ 4 & 1 & -2 \end{bmatrix}$$
$$= 1(-6+0) + (4+0) + 0(2+12)$$
$$= -6 + 16 = 10$$

$$I_1 = \frac{\Delta 1}{\Delta} = \frac{110}{230} = 0.4782\text{A}$$
$$I_2 = \frac{\Delta 2}{\Delta} = \frac{40}{230} = 0.174\text{A}$$
$$I_3 = \frac{\Delta 3}{\Delta} = \frac{10}{230} = 0.0434\text{A}$$

Current through 1Ω resistor I_1 = 0.4782A

Current through 2Ω resistor $(I_1 - I_3)$ = 0.4782 – 0.043 = 0.4352A

Current through 3Ω resistor = $(I_2 + I_3)$ = 0.174 + 0.043 = 0.217A

Current through 4Ω resistor I_2 = 0.174A

Current through battery $(I_1 + I_2)$ = 0.4782 + 0.174 = 0.6522A

Example.2

In a house, the following equipment are used.

1) 1HP motor works 3 hours per day.
2) 40w, 10 lamps glow 5 hours per day.
3 900w capacity of heater works 2 hours per day.

Calculate the cost of energy consumed in 60 days.

[For domestic purpose unit rate is (1–100)-NIL,(101–200) Rs.3.50 (201–500) = Rs. 4.60, > 500 units = Rs. 6.60

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